

Emanuele Smisi

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MSc student in Computer Science & AI at Sapienza with a background in Economics and Big Data, building data pipelines, ML models and distributed systems. Looking for a **curricular internship for my Master's thesis** in machine learning / data systems, available from **spring 2027** for 3-6 months.

Education

Sapienza University of Rome

Rome, IT

MSc in Engineering in Computer Science and Artificial Intelligence

Sept 2025 - Exp. 2027

- **Coursework:** Machine Learning, Reinforcement Learning, Artificial Intelligence, Data Management, Algorithm Design, Dependable Distributed Systems, Mobile Edge & Cloud Computing, Economics and Computation, Formal Methods, Interactive Graphics, Cybersecurity

Università degli Studi Roma Tre

Rome, IT

BSc in Economics and Big Data (Final grade: 99/110)

Sept 2022 - Oct 2025

- **Thesis:** Beyond Market Data: Reddit Sentiment for Hourly Bitcoin Forecasting (VADER vs FinBERT)

Projects

E.C.H.O.: Distributed Real-Time Seismic Monitoring

[GitHub](#) 

Sapienza, Advanced Programming · team of 5

- Microservice platform (**FastAPI**, **WebSockets**, **Docker**, **PostgreSQL**, **React**) that ingests 20 Hz sensor streams and classifies earthquakes, explosions and nuclear-like events in real time.
- Owned the **processing service**: per-sensor sliding window + **FFT** dominant-frequency analysis, event classification and severity score; unit tests in **CI**.

The Purchasing Power of the Tech Worker: Data Warehouse

[GitHub](#) 

Sapienza, Data Management · team of 2

- Integrated 3 heterogeneous sources (65k-response Stack Overflow survey, ISO country master, cost-of-living index) into a **PostgreSQL** star schema with an idempotent **Python ETL**.
- Answered *where a tech salary buys the most* with **OLAP queries** (ROLLUP, window functions, medians) and a Streamlit dashboard.

BSc Thesis: Reddit Sentiment for Bitcoin Forecasting

[GitHub](#) 

Università Roma Tre

- Built a leakage-safe pipeline on 17.5k hourly BTC observations and ~85k Reddit texts, comparing **VADER** and **FinBERT** sentiment across ARIMA, LSTM and **XGBoost** models.
- Assessed gains with **Diebold-Mariano tests** and Model Confidence Sets: sentiment helps only modestly (best: XGBoost + VADER at 4h), and naïve baselines remain hard to beat.

Machine Learning in Reduced Feature Spaces

[GitHub](#) 

Sapienza, Machine Learning · team of 2

- Trained a **CNN (PyTorch)**, an FNN and **K-Means** on image, price and customer data, comparing **PCA vs autoencoder** compression: R^2 0.94 $\rightarrow$ 0.92/0.91 with far less compute.

MyPiggyBank: User-Centred Design of a Budgeting App

[GitHub](#) 

Sapienza, User-Driven Software Engineering · team of 5

- Ran a 31-respondent user survey, derived requirements, modelled the dialogue with state-transition networks and built a React prototype.

Experience

IT Consultant (pro bono)

Rome, IT

Bespoke tailoring business · 2 locations

Jun 2026 - Jul 2026

- Ported an early-2000s **order-management web app** (Classic ASP/VBScript + MS Access, 25 years of order history) stuck on a Windows XP VM, to **Windows 10/11** and modern browsers.
- Migrated the database from Access to **SQL Server 2022 Express**: schema conversion, row-count-validated data transfer, fixes for date formats, field lengths and 32/64-bit ODBC drivers.
- Built an **automated migration kit** (pre-checks, DB setup, data migration, verification) that cut setup to ~1-2 hours per machine, **rolled out to 3 workstations in 2 locations**, with nightly cloud backups.

Waiter (part-time), Restaurant, Rome, alongside full-time studies

May 2023 - Present

Skills

Programming: Python, SQL · **Languages:** Italian (Native), English (B2)

ML & Data Science: scikit-learn, PyTorch, TensorFlow/Keras, XGBoost, Hugging Face Transformers, pandas, NumPy

Data Systems: PostgreSQL, SQL Server, data warehousing (star schema, ETL, OLAP)

Tools: Git, Docker, FastAPI, AI-assisted development (Claude Code)

Interests: Volleyball (6 years, team), football, gym